

CONDITIONAL IMPERFECT MAINTENANCE OF A DETERIORATION PROCESS

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We consider a gamma process for degradation modeling of a system, which is periodically inspected. At each inspection, a decision is taken in respect to the level of the degradation process. We consider the following maintenance framework:

- The system degradation indicator is more related to a service quality than a health indicator and the system failure is not considered in this paper;
- A perfect preventive maintenance is performed if the degradation level exceeds a fixed safety threshold L : the operation leads to a "as good as new" system;
- An imperfect preventive maintenance is performed if the degradation level is between L and a preventive threshold M : the imperfect action is modelling to an arithmetic reduction of degradation of order 1. In this later case, the improvement is proportional to the degradation level at the inspection time (with a reduction factor ρ with $0 \leq \rho \leq 1$);
- No action is performed if the degradation level is lower than M ;
- The safety threshold exceeding is not self-announced and this event is detected only during an inspection;
- Maintenance actions do not induce a delay.

Considering the cost of the inspections, of the perfect and imperfect preventive maintenances, we derive the closed form expression of the average long run cost of the maintenance policy in order to optimize the expected total cost between two CM (thanks to the renewal theory). A sensitivity analysis is performed and the robustness of the optimal parameters with respect to uncertainty.

Keywords: Imperfect Maintenance, Gamma Process, Maintenance Optimization, Degradation Process.

References

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